

NGWENYA MPHO

Electrical Engineering (Computer Systems) Student

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PROFFESION SUMMARY

Final-year Electrical Engineering (Computer Systems) student at Vaal University of Technology, with practical experience in embedded systems design, sensor integration, and microcontroller-based control. Contributed to an ESP32-based smart irrigation monitoring and control system project. Completed coursework spanning electronics, digital systems, networks, operating systems, and software engineering. Seeking a work-integrated learning placement to fulfil national diploma requirements.

EDUCATION

National Diploma in Electrical Engineering – Computer Systems

Expected 2026

Vaal University of Technology (VUT)

- Core Courses: Electrical Engineering, Electronics, Digital Systems, Networks, Engineering Programming, Operating Systems, Software Engineering
- Academic Average: 70%

National Senior Certificate

2019

Kenneth Masekela Secondary School

- Distinctions: Physical Science, Life Sciences
- Academic Average: 74%

PROJECTS

Final Year Project: ESP32-Based Smart Irrigation, Monitoring and Control System in Farms

2026

Vaal University of Technology (VUT)

- Designed a sensor network across five ESP32 nodes (DHT22, soil moisture, PIR, rainfall, and ultrasonic sensors) to capture real-time environmental data on the farm.
- implemented a programmed logic that makes irrigation decision in order to automate the irrigation process in the farm.
- Developed a web-based server to monitor and control the system in real-time.
- Improved the water efficiency in farms.

Second Year Project: Music Amplifier from Cell Phone to Speaker

2025

Vaal University of Technology (VUT)

- Captured audio signal from a cell phone using aux cable.
- Amplified the signal using a LM386 transistor circuit.
- Delivered the amplified signal to a speaker.
- Provide control over the speaker audio level using potentiometer.

SKILLS

Programming: Python | C++ | C | Assembly | HTML & CSS | SQL | Bash | JavaScript

Framework: Flask

Data Science: Python for Data Science | Matplotlib | Pandas

Cloud Computing: OpenStack | Proxmox | Amazon Web Service (AWS)

Database: MySQL | SQLite

Software: MATLAB | Multisim | Arduino IDE | Cisco Packet Tracer | MS Office Suite

Hardware & Embedded Systems: Microcontrollers | IoT Design | Circuit Design | Sensor Integration